

Supplementary Material

A high-resolution, process-based PFAS surface-water fate-and-transport model in SWAT⁺: engine development, cross-validation, and a source-attribution demonstration on the Rogue River, Michigan

Vahid Rafiei

Contents

S1 Parameters and variables	1
S1.1 PFAS physico-chemical and sorption parameters	1
S1.2 Calibration parameters	1
S1.3 Uncertainty-analysis design	1
S2 Per-reach calibration data	2
S3 Engine implementation	2
S3.1 Source-file inventory	2
S3.2 Soil three-phase equilibrium solver	3
S3.3 Land-phase mobilization	4
S3.4 In-stream routing	4
S4 Input and output formats	4
S5 Build, activation, and cross-validation	5

This document provides (i) the variables and parameters used for calibration and uncertainty analysis, (ii) the per-reach calibration data, and (iii) the technical implementation detail needed to reproduce or extend the engine: source-file inventory, governing equations and solver, input/output file formats, build and activation procedure, and the cross-validation harness. Section and table numbers are prefixed “S”. It accompanies the main text and uses the same notation.

S1 Parameters and variables

S1.1 PFAS physico-chemical and sorption parameters

Table S1 lists the compound and sorption parameters the engine reads. The global compound record (pfas.dat) carries the molecular weight, aqueous solubility, the Langmuir

air–water–interface coefficients, and the runoff/percolation partition; the per-HRU initialization (`pfas_hru.ini`) carries the initial soil pool and the Freundlich coefficients, assigned by soil class and land use. Values shown are the PFOS defaults used in this study.

Table S1: PFAS fate parameters read by the engine, with the PFOS values used here. C_w is the aqueous concentration; symbols match Eq. (1) of the main text.

Symbol	Description	Source / file	PFOS value
M_w	molecular weight (kg mol^{-1})	<code>pfas.dat</code>	0.50013
S	aqueous solubility (mg L^{-1})	<code>pfas.dat</code>	680
K_L	Langmuir AWI affinity ($\text{L } \mu\text{mol}^{-1}$)	<code>pfas.dat</code>	0.137
$\Gamma_{\max}$	Langmuir AWI max surface excess ($\mu\text{mol m}^{-2}$)	<code>pfas.dat</code>	2500
f_{perc}	runoff/percolation partition (–)	<code>pfas.dat</code>	0.20
<code>sol_pfas</code>	initial soil PFOS pool ($\mu\text{g ha}^{-1}$)	<code>pfas_hru.ini</code>	per HRU
k_f	Freundlich coefficient ($(\text{nmol kg}^{-1})/(\text{nM})^n$)	<code>pfas_hru.ini</code>	216–450
n	Freundlich exponent (–)	<code>pfas_hru.ini</code>	0.33–0.48
<code>enr</code>	sediment enrichment ratio (–)	<code>pfas_hru.ini</code>	per HRU
A_{aw}	air–water interfacial area ($\text{m}^2 \text{m}^{-3}$)	from θ , d_{50}	derived
k_d	in-stream sediment–water partition (L kg^{-1})	$K_{oc} \cdot f_{oc}$	derived

S1.2 Calibration parameters

Calibration adjusts two global multipliers in `pfas_calib.dat`: a soil-loading scale on every HRU’s initial soil pool and a multiplier on the in-stream organic-carbon partition. Because in-stream concentration is, to first order, linear in the soil-loading scale, the magnitude calibration is nearly analytic (one forward evaluation fixes the scale to first order). The calibrated values are soil-loading = 0.11 and K_{oc} multiplier = 1.0.

S1.3 Uncertainty-analysis design

Table S2 gives the five-parameter Latin-hypercube design (120 members, uniform priors), the calibrated value of each parameter, and its Spearman rank correlation with the reach-mean in-stream PFOS concentration across the ensemble. The soil-loading multiplier dominates the response ($\rho = 0.95$); the remaining parameters are second-order or negligible, consistent with the near-linear structure of the problem.

Table S2: Uncertainty-analysis parameter design: prior ranges, calibrated values, and Spearman rank correlation with reach-mean in-stream PFOS ($n = 120$ members).

Parameter	Prior range	Calibrated	Spearman ρ
Soil-loading multiplier	[0.05, 0.22]	0.11	+0.95
In-stream K_{oc} multiplier	[0.5, 2]	1	-0.11
Langmuir affinity K_L	[0.07, 0.27]	0.137	+0.08
Langmuir max $\Gamma_{\max}$	[1500, 3500]	2500	-0.22
Runoff/percolation partition	[0.1, 0.4]	0.2	+0.07

S2 Per-reach calibration data

Table S3 reports, for each of the 20 gauged reaches, the station-maximum observed PFOS, the calibrated modeled concentration (soil-loading = 0.11, no point source), and the 5th–95th-percentile predictive envelope from the 120-member ensemble. “M” marks the seven Rogue mainstem reaches (stream order ≥ 4); “T” marks the thirteen tributary reaches. The envelope brackets the observation on six of seven mainstem reaches and almost no tributary reaches—the basis for the P -factor decomposition in the main text.

Table S3: Per-reach observed, modeled, and 5–95% predictive-envelope PFOS (ng L^{-1}). Class M = mainstem (order ≥ 4), T = tributary.

Reach	Class	Obs	Modeled	Env. 5%	Env. 95%
2	M	27.0	15.4	9.8	25.5
1	M	19.0	14.7	9.3	24.3
10	M	9.6	12.4	7.9	20.7
15	M	7.2	9.2	5.8	15.4
18	M	6.8	7.7	4.8	12.9
11	M	6.7	11.2	7.1	18.7
26	M	6.3	6.2	3.9	10.4
290	T	24.0	5.4	3.4	9.1
519	T	14.0	9.7	6.2	15.8
449	T	6.0	0.8	0.4	1.4
348	T	4.7	50.8	32.6	82.4
488	T	4.6	0.1	0.0	0.2
224	T	3.8	25.3	16.3	40.9
321	T	3.6	7.6	4.8	14.5
195	T	3.4	20.7	13.2	35.0
157	T	3.3	1.2	0.7	2.2
96	T	2.6	13.9	8.7	23.3
112	T	2.5	32.8	21.2	54.8
235	T	1.9	14.8	9.5	24.2
81	T	1.5	21.9	13.8	35.4

S3 Engine implementation

S3.1 Source-file inventory

The engine is confined to a small set of new free-form Fortran routines (Table S4) plus minimal hooks in existing SWAT⁺ files (`hru_hyds`, `ru_control`, `res_control`, `command`, `sd_channel_control3`, `constituent_mass_module`, `main`). The stock pesticide initialization is untouched.

S3.2 Soil three-phase equilibrium solver

For each HRU and soil layer the engine partitions the total PFAS mass M among a dissolved, a Freundlich solid-sorbed, and a Langmuir air–water-interfacial phase by solving, for the

Table S4: New PFAS engine source files and their roles.

File	Role
<code>pfas_module.f90</code>	derived types (compound database, soil/channel pools), parameters
<code>pfas_read.f90</code>	read <code>pfas.dat</code> (global database) and <code>pfas_hru.ini</code> (per-HRU init)
<code>pfas_count_read.f90</code>	early constituent count to self-activate the hydrograph infrastructure
<code>pfas_lch.f90</code>	land phase: soil three-phase equilibrium + runoff/lateral/leach losses
<code>pfas_sed.f90</code>	sediment-bound (enrichment-ratio) loss
<code>pfas_output[_module].f90</code>	land-phase per-HRU mass-balance output
<code>pfas_cha[_module].f90</code>	in-stream linear-partition routing + benthic exchange
<code>pfas_cha_read.f90</code>	channel initialization + optional point-source read
<code>cha_pfas_output.f90</code>	daily per-reach channel output (<code>channel_pfas.day.txt</code>)
<code>pfas_xval_selftest.f90</code>	standalone cross-validation against <code>ch_rtpest</code>

aqueous concentration $C_w > 0$,

$$M = \theta V C_w + m_s k_f C_w^n + \Gamma_{\max} \frac{K_L C_w}{1 + K_L C_w} A_{aw}. \quad (1)$$

The right-hand side is strictly increasing in C_w , so the root is unique. The solver brackets the root, then refines with an Improved-Halley iteration retaining the bisection bracket as a guaranteed fallback (Algorithm 1). Against a 40-digit decimal reference the double-precision solver holds a worst-case relative error of 1.7×10^{-7} over 29,160 stiff parameter combinations.

Listing 1: Soil-equilibrium root solve (pseudocode).

```
f(Cw) = theta*V*Cw + m_s*kf*Cw^n + Gmax*KL*Cw/(1+KL*Cw)*A_aw - M
# 1. bracket: lo = 0, hi = M/(theta*V) (aqueous-only upper bound)
# 2. bisection until f changes sign within a safe bracket
# 3. Improved-Halley refinement from the bracket midpoint:
# Cw <- Cw - (2 f f') / (2 f'^2 - f f'')
# accept only if it stays inside the bracket; else bisect
# 4. converge on |f|/M < tol; back-substitute dissolved/solid/AWI masses
```

S3.3 Land-phase mobilization

Dissolved PFAS leaves the surface layer with surface runoff and percolation through the runoff/percolation partition f_{perc} , deeper layers with lateral flow, and the solid-sorbed fraction with sediment yield through the enrichment ratio, exactly paralleling the SWAT⁺ pesticide land phase. The four daily losses are accumulated per HRU and per compound and become the constituent loads delivered to the channel network; the soil pool decrements consistently. A whole-run land-phase balance closes to a relative residual of order 10^{-5} .

S3.4 In-stream routing

In the channel the dissolved fraction is $f_{\text{sol}} = 1/(1 + k_d c_s)$, with k_d the partition coefficient (scaled by the calibrated K_{oc} multiplier and reach bed-carbon fraction) and c_s the suspended-sediment concentration. The sorbed fraction settles to the bed; bed PFAS resuspends, diffuses across the sediment–water interface, and is permanently lost by burial; dissolved

and sorbed fractions leave the reach in outflow. These are exactly the non-reactive terms of the SWAT⁺ pesticide channel solver (**ch_rtpest**) with aqueous decay, volatilization, and metabolite production removed. Reservoirs are mass-conserving completely mixed cells; point sources are per-reach daily loads added to reach inflow before routing.

S4 Input and output formats

pfas.dat (global compound database). Two header lines, then one list-directed record per compound and a terminating 0 END record:

```
id name mw sol kl lm percop
1 PFOS 0.50013 680.0 0.137 2500.0 0.20
0 END 0.0 0.0 0.0 0.0 0.0
```

pfas_hru.ini (per-HRU soil initialization). Per HRU: a hru <id> nly <n> header, the constituent count, the per-layer d_{50} , then, per compound, four list-directed records giving the initial soil pool (**sol_pfas**, $\mu\text{g ha}^{-1}$), k_f , n , and the enrichment ratio, one value per layer.

pfas_calib.dat (calibration multipliers). A single line: **soil_scale koc_scale** (e.g. 0.11 1.0).

Point-source file (optional). Per-reach daily PFAS load (kg d^{-1}) added to reach inflow; the injected mass is tracked in the in-stream balance.

channel_pfas_day.txt (daily per-reach output). Columns (Table S5); the final column, **conc_ngL**, is the quantity compared against ambient observations.

Table S5: Daily per-reach PFAS output columns.

Column	Meaning
jday mon day yr	date
unit	channel (reach) number
gis_id name	GIS id and reach name
pfas	compound index
tot_in_kg	total PFAS inflow to the reach
sol_out_kg	dissolved mass leaving in outflow
sor_out_kg	sediment-sorbed mass leaving in outflow
settle_kg	mass settled to the bed
resus_kg	mass resuspended from the bed
diffuse_kg	sediment-water diffusive exchange
bury_kg	mass permanently buried
water_kg	water-column PFAS storage
benthic_kg	benthic (bed) PFAS storage
conc_ngL	outflow concentration (ng L^{-1})

S5 Build, activation, and cross-validation

Build. The engine compiles with the Intel `ifx` compiler and NetCDF (`ifx -O3 -ipo; build_pfas_engine.sh`). New source files require a clean build because the build globs source at configure time.

Self-activation. Routing PFAS through a real network needs the constituent-hydrograph infrastructure that SWAT⁺ allocates only when a constituent is declared. `pfas_count_read` runs early and increments the total constituent count (`cs_db%num_tot`) before `hyd_connect`, so the infrastructure is allocated whenever `pfas.dat` is present; a PFAS run therefore needs no dummy constituent. Aggregation in `ru_control` carries the PFAS pool through the HRU→routing-unit step (mirroring the pesticide pool), and gating on the scalar `cs_db%num_tot` avoids touching unallocated hydrographs.

Cross-validation. `pfas_xval_selftest` sets up one synthetic reach and drives both `pfas_cha` and the stock `ch_rtpest` (configured with unit decay, zero volatilization, no metabolites—its linear-partition subset) with identical inflow and parameters over a 60-day sweep including a dry-reach day. The two routines agree on all nine shared mass terms to a maximum absolute difference of 2.4×10^{-7} kg and a maximum relative difference of 2.6×10^{-7} , establishing that the in-stream mathematics is reused rather than reinvented and is not a source of error.

Uncertainty-ensemble execution. The 120-member Latin-hypercube ensemble (Sec. S1) was run as 120 independent forward simulations, each in an isolated working directory, in parallel on a single 32-vCPU cloud instance (AWS EC2 `c7i.8xlarge`, spot). The full ensemble completed in 14.4 min of compute (15 min 05 s end to end including provisioning and result transfer), a mean of 172 s per member.